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Human vision inspired feature extraction for facial expression recognition

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Abstract

Facial expression is a powerful way for human emotional communications. According to various applications, automatic facial expression recognition becomes an interesting problem for researchers in different areas. An automatic facial expression recognition system recognizes the expressed emotion in input facial image using several processing stages. Feature extraction is a vital step in facial expression recognition. One of the most widely used techniques for feature extraction in machine vision is utilizing Gabor filter which is sensitive to lines at various orientations. The disadvantage of Gabor filters is their computational cost and large feature vector length. Inspiring the human vision system and stimulations of complex cells, in this paper, firstly, facial image is convolved with Gabor filters. Then, the achieved convolution matrices are properly coded based on the maximum and minimum responses. Finally, the feature vector is obtained by calculating the histogram of these codes. The length of achieved histogram for 16 and 8 Gabor filters are 240 and 56, respectively, which is considerably less than keeping all Gabor responses. The proposed method is (person-independently) evaluated on four facial expression recognition datasets including CK+, SFEW, MMI, and RAF-DB. The experimental results show that the proposed method outperforms existing image texture descriptors in facial expression recognition in both controlled and uncontrolled images.

Keywords Facial expression recognition · Complex cell · Human vision system · Gabor filter · Image texture description

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1 Introduction

Facial expression influenced by emotional state of individuals is one of the powerful non-verbal human communications for all cultures. Facial appearance and the shapes of face components, such as mouth and eyebrows, represent the human personality which can be used in physiognomy [50]. According to psychological researches, there are six basic emotions with similar facial expressions across different cultures; these emotions are anger, disgust, fear, happiness, sadness, and surprise [10]. Expressing these emotions in the face contains important information about human reaction in the environment. During these basic emotions, some physiological changes are occurred in human body. For example, disgust emotion increases the size of right-frontal asymmetry in the central nervous system; on the other hand, happiness one increases the size of left-frontal asymmetry in this system [12, 43]. Psychophysiological experiments on autonomic activity of emotional facial configurations show that these basic emotions change finger temperature, skin conductance, and heart rate, in human body [32]. Anger emotion leads to a blood flow increase through the hands; this blood flow increasing makes person ready for battle [32]. According to the importance of facial expression in different problems and applications, automatic facial expression recognition has attracted many behavioral science, neurology, artificial intelligence, and machine vision researchers. Consequently, in this paper, an automatic facial expression recognition method is presented.

There are various applications for automatic facial expression recognition and analysis systems in different areas of engineering, medicine, psychological sciences, etc. Human-computer interaction, intelligent cameras, and emotional robots are some usages of facial expression recognition in engineering. Moreover, facial expression recognition is employed in medicine for pain detection in patients [23, 26, 40]. In [23], the authors combine RGB, depth, and thermal information of facial videos and employed a spatiotemporal filter to detect paint and its intensity. Since the severity of depression is dependent to human nonverbal behavior [16], specially facial expression, one of the most important applications of automatic facial expression analysis can be utilized in psychological for diagnosis of depression [25, 52].

In general, three main processing steps of facial expression recognition are face detection, feature extraction, and classification. In face detection, the positions of face and its components in the input image are estimated. Calculation of important facial expression information from facial image to create feature vector is feature extraction step. Finally, determining the type of expressed emotion in the face by using the extracted feature vector is called classification.

A proper feature extraction plays important role to improve the facial expression recognition accuracy; two main groups of feature extraction methods, including geometric and appearance based, are defined in facial expression recognition [56]. To extract geometric features, locations of facial components are first detected and represented via a number of feature points (landmarks) or contours. Next, a feature vector is calculated using the detected (x, y) coordinates. This feature vector, containing geometric information of facial components, is calculated based on the distances of landmark points, slope of their connected lines, the arrangement of points (the way points are placed next to each other), etc. Conversely, in appearance feature extraction, without the need for face points, only the gray level values of the pixels along with the relationship of each pixel with a set of neighboring pixels are used to represent texture information of facial image. Appearance feature extraction is typically performed via different image filters or texture descriptors. In conclusion, it can be said that geometric features represent the shape and position of facial components such as mouth, eyes,

and forehead. On the other hand, appearance features contain various skin texture information such as creases, furrows, lines, and facial wrinkles [56].

As an example of geometric feature-based methods, in [29, 30], a face geometric model consisting of specific distances of 30 facial points is defined for facial expression recognition. In [46], some basic facial expressions are classified based on correlation functions of such facial geometric features as the position of lips corners related to their center, mouth position, and the height and width ratio of mouth. In another work, geometric feature vectors are extracted based on the shapes of mouth, eyes, and eyebrows in face [41]. To achieve these features, the distances of some specific facial points (i.e. the distance between eyebrows and the center of eyes) are compared between the neutral image and emotional ones. Finally, an advanced Kohonen self-constructed map (KSOM) method is used to classify the extracted geometric features. In one research, facial landmark points are localized utilizing the extended ASM algorithm; then, various geometric features are defined based on the length and slope of lines, the area of different regions, etc. Finally, different classifiers, such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), are used to classify the extracted features [47]. In [14], geometric features have been extracted from video data using Elastic Bunch Graph Matching landmark detection along with Kanade-Lucas-Tomaci (KLT) tracker on landmark points, lines, and triangles for facial expression recognition. In [1], a geometric feature extraction approach using area and angels of triangles based on facial landmark points is introduced for facial expression recognition in video sequences. Hsu et al. proposed a bag of geometric distance-based features codebook via AAM facial landmark detection for facial expression recognition using SVM classification [21].

In order to describe the texture of images in appearance feature-based method, different image filters are applied to facial images. In this context, Gabor filters (or wavelets) [6] have attracted much attention due to their sufficiency [2, 17, 59, 61]. However, the disadvantage of Gabor filter bank is its high computational cost. In the next years, Local Binary Patterns (LBP) [45] is introduced as another image texture descriptor. Different variations of LBP are employed for appearance-based facial representation [18, 22, 31, 60]. In [54], boosted version of LBP descriptor (Boosted-LBP) and different classifiers are used for facial expression recognition. Boosted LBP extracts discriminative LBP features from facial images to improve the performance of facial expression recognition. In [62, 63], in case of video-based data and dynamic image texture description, Volume Local Binary Patterns (VLBP) and Local Binary Patterns from Three Orthogonal Planes (LBP-TOP) are introduced and employed in facial expression recognition. Jia et al. [24] have employed LBP and LBP-TOP appearance features and singular value decomposition in a macro-to-micro transformation model for micro-expression recognition in small dataset. Liu et al. presented Boosting-POOF algorithm which extracts HOG features around various facial landmarks as local mid-level features and uses a SVM based Adaboost feature selection for facial expression recognition in uncontrolled conditions [37]. In [51], LBP was applied to emotional and neutral difference image in a semi-dynamic facial expression recognition method. Histogram of Oriented Gradients (HOG) [55], Patterns of Oriented Edge Magnitudes (POEM) [58], Local Directional Number pattern (LDN) [49], Temporal POEM (TPOEM) [5] are other appearance based features utilized for facial image description. Nigam el al. extracted HOG features from discrete wavelet transform (DWT) domain for facial expression recognition [44]. Li et al. proposed a set of salient patterns at facial landmark locations and utilized appearance-based LBP and geometric features along with SVM classifier [34]. Ghimire et al. divided facial images based on landmark points into local regions and combined geometric and appearance NCM and LBP features for facial expression recognition using SVM classifier [15]. Hsieh et al. utilized

ASM algorithm for facial landmark detection and employed such operation as Gabor filters for appearance-based facial feature extraction [20]. Multi-Directional Multi-Level Dual-Cross Patterns (MDML-DCPs) was introduced as a face texture descriptor for face recognition under different illuminations, expressions, and head poses [8]. Ben et al. developed DCP and Hot Wheel Patterns (HWP) as extended version of DCP for video-based binary description and proposed DCP from three orthogonal planes (DCP-TOP) and HWP from three orthogonal planes (HWP-TOP) in micro-expression recognition [3]. On the other hand, there are various learnable descriptors proposed for facial image analysis [9, 38]. Compact Binary Face Descriptor (CBFD) is a feature learning approach to calculate histogram of binary codes computed based on pixel difference vectors (PDVs) using a clustering algorithm [38]. Context-aware local binary feature learning (CA-LBFL) is another face descriptor proposed for face representation [9].

An important weakness of geometric feature-based methods is their dependency on the accuracy of facial components detection. In other words, these methods are strongly influenced by facial landmark localization error. In contrast, appearance feature-based methods use image filtering and texture description and are more robust against this error. Therefore, in this paper, appearance features are employed and an appearance feature extraction method is proposed for facial expression recognition. Since the geometric displacements of facial components appropriately affect the proposed feature, it can be said that the extracted feature contains some geometric properties of the face. However, it does not directly use the location of facial components; therefore, in contrast to landmark-based geometric feature extraction, the proposed method is more robust against landmark localization error. As mentioned earlier, Gabor filters have attracted much attention in facial expression recognition. However, the disadvantage of Gabor filters in feature extraction is their large feature length. For example, the feature vector length for a 150×110 pixels image using 16 Gabor filters is reached 264,000 ($150 \times 110 \times 16$). This feature vector requires a large amount of memory to store and affects the computational cost of classification stage. In this paper, a novel feature extraction method is introduced to code Gabor filter convolution coefficients of each pixel via a single number to reduce feature dimensionality. In the proposed method, inspired by the human vision system, the length of feature vector for each block of image is 240; therefore, by a proper description of the Gabor coefficients, the length of final feature vector is considerably reduced. The proposed feature extraction method has superior performance compared to the existing image descriptor in facial expression recognition.

The rest of this paper is organized as follows: The proposed facial expression recognition system is described in the next section. In Section 3, the utilized facial expression datasets are described and the proposed method is evaluated. Finally, Section 4 concluded this paper.

2 Proposed facial expression recognition system

The overview of the proposed facial expression recognition system is described in Fig. 1. According to this figure, the proposed system comprises three vital steps including (1) preprocessing, (2) feature extraction, and (3) classification. As mentioned before, the focus of this study is based on feature extraction step, which the detailed process of this step is shown in this figure. Briefly, in feature extraction step, Gabor filters are convolved with the normalized image. Then, the results of Gabor filters convolution are coded. Finally, the coded matrix is divided into several blocks and the histograms of these blocks are calculated to create final feature vector. Three steps of this block diagram are described in the rest of this section.

2.1 Preprocessing

In order to extract facial features from input image, it is necessary to separate the face area from background and scale it into an appropriate size. For this reason, first, face and its landmark points are localized based on the method presented in [28]. Next, face region is cropped from original image and in-plane head rotation is removed based on the locations of two eyes. Finally, facial image is normalized into 150×110 pixels and is converted into grayscale image. An example of this process is show in Fig. 2.

2.2 Feature extraction

Facial image texture description is an applicable feature extraction approach in facial expression recognition. In this section, a novel appearance feature extraction (or texture description) method, inspired by human vision system, is proposed. For this purpose, a short introduction of the human vision system and its relation with the proposed method are discussed. Then, Gabor filters and the proposed feature extraction method based on these filters and human vision system is presented.

In the human vision system, by viewing a line at a certain orientation and magnitude, associated complex cells are stimulated, while the other ones in different orientations and magnitudes are not stimulated. Neurons of lateral geniculate nucleus have circular receptive fields. These neurons are sensitive to edges\lines light differences; however, they could not detect the orientations of edges\lines. In the brain visual cortex, neurons respond to a range of orientations. Consequently, different neurons stimulated via line in different orientations. Each neuron in visual cortex, called a complex cell, consists of the adjacency of several neurons in geniculate nucleus; this is shown in Fig. 3 [27].

Gabor filter is designed based on human vision system in machine vision. However, it leads to a large feature vector. In this paper, inspiring human vision system and vision cells-stimulation, a new feature extraction method is proposed. In this method, instead of all Gabor filters, only those filters directly related to the information of input image (equivalent to stimulated neurons) are considered. Thus, the dimension of the extracted feature vector is reduced. In the rest of this section, Gabor filters and the proposed feature extraction based on these filters are introduced.

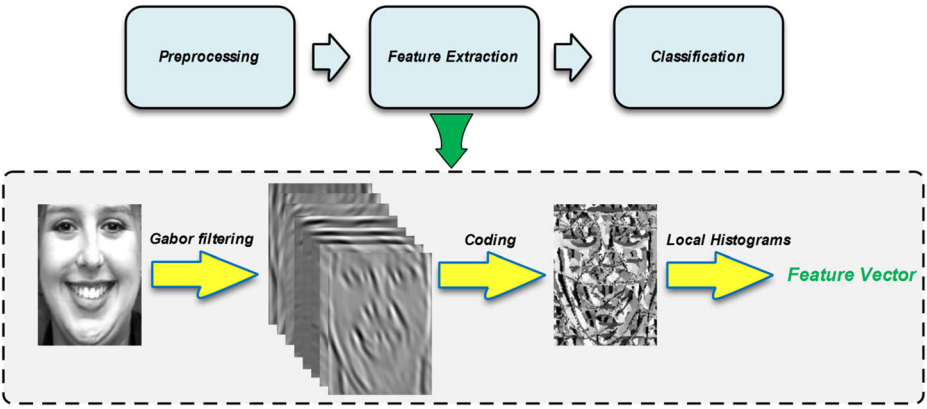

Fig. 1 Overview of the proposed facial expression recognition system

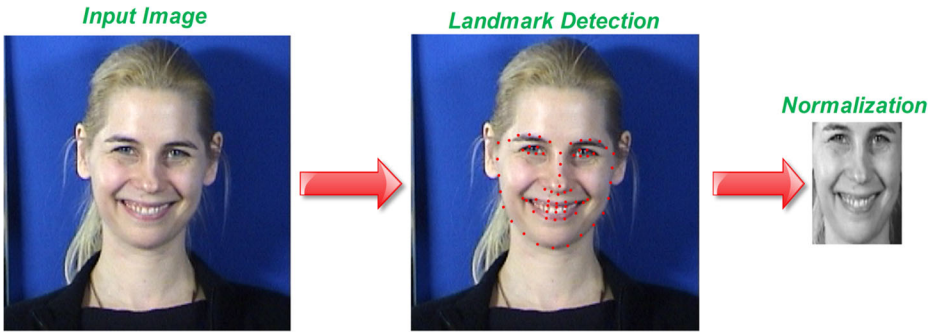

Fig. 2 Face localization and normalization

2.2.1 Gabor filters

Gabor filters [6] are very similar to the human vision system and can describe lines at various orientation; hence, these filters can be utilized to model visual cells [13, 53]. The mathematical formulation of these filters is defined as:

$$\begin{aligned} \psi_j\left(\vec{x}\right) &= \frac{k_j^2}{\sigma^2}\exp\left(-\frac{k_j^2 x^2}{2\sigma^2}\right)\left[\exp\left(\mathrm{i}\,\vec{\mathrm{k}}_j\,\vec{x}\right)-\exp\left(-\frac{\sigma^2}{2}\right)\right] \\ \vec{k}_j &= \begin{pmatrix} k_{jx} \\ k_{jv} \end{pmatrix} = \begin{pmatrix} k_v\cos\phi_u \\ k_v\sin\phi_u \end{pmatrix}, k_v = 2^{\frac{v+2}{2}}\pi, \phi_u = \mu\frac{\pi}{8} \end{aligned} \tag{1}$$

where, μ is the orientation of filters (0–7), ν represents the size (or frequency) of filters (0–4), $\vec{x}$ indicates the pixel coordinate in the image (x, y), and σ Gaussian function parameter is set to 2π.

Two dimensional Gabor filters consist of complex values and are convolved with input image to extract features. Real parts of Gabor filter for 5 scales/sizes and 8 orientations are shown in Fig. 4. Let's assume that each Gabor filter simulates a complex cell in the human

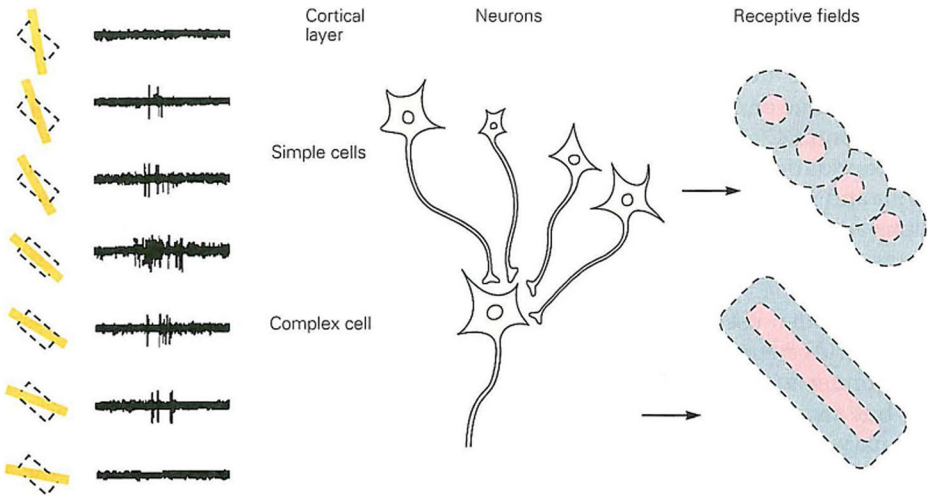

Fig. 3 The responses of complex cell in the visual cortex to light line in different orientations [27]

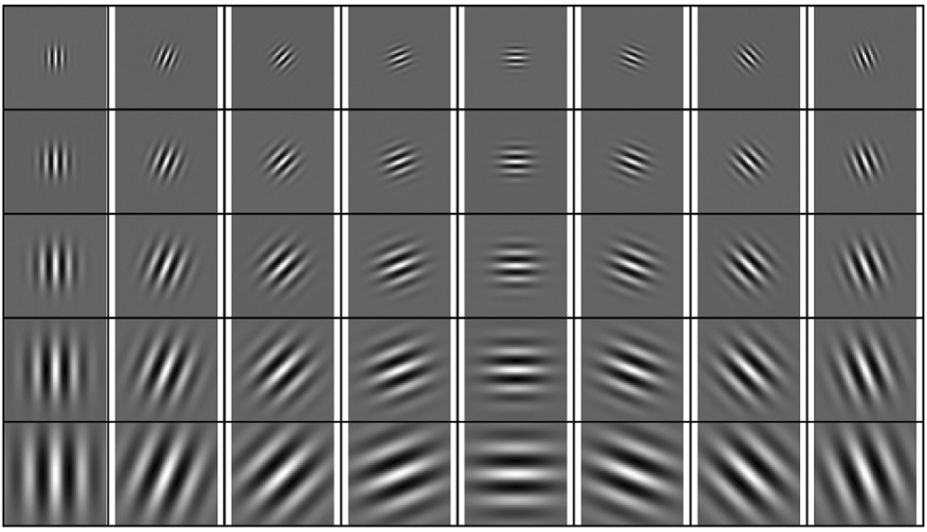

Fig. 4 Real parts of Gabor filters for 5 scales and 8 orientations

vision system [53]; the convolution result of the filter and image indicates stimulation/not-stimulation of the corresponding complex cell. In the conventional Gabor filter-based image description methods, the image feature vector usually contains outputs of all Gabor filters. However, it leads to a large feature vector; because of this, the computational complexity in feature classification stage is increased.

2.2.2 Gabor filter coding

Convolving Gabor filters and the image produces a large feature vector requiring a large amount of memory to store and increasing the computational cost of classification stage. In this section, a novel feature extraction method is proposed achieving a short feature vector instead of directly using all convolution results of Gabor filters.

If convolution output for a particular Gabor filter and input image in a specific region is high, then image information in this region is proportional to this filter. In other words, this filter (the same as a complex cell) is stimulated. Considering the outputs of all Gabor filters in a particular region of image, the filter with the maximum response is the most similar filter to this region; while, the filter with minimum response has minimum similarity with the region. In this study, utilizing this convolution property, the filters with the maximum and minimum responses are considered as the complex cells with maximum and minimum stimulations, respectively.

The information of two selected filters with the maximum and minimum responses should be utilized to form feature vector. Concatenating the responses of these filters creates a long feature vector. To create a low dimensional facial feature vector, this paper proposes a new coding method for responses of the selected filters. These filters are coded using a set of codes which can be used to form a histogram-based feature vector. To understand the feature extraction method, here is an example:

Let's consider 16 Gabor filters in 2 sizes and 8 orientations. In the first step, these filters are convolved with facial image. The result of convolution is 16 complex matrices with the same

dimension as input image; in other words, for each pixel of the image, there are 16 complex values corresponding to 16 Gabor filters. In each pixel, the indices of filters with maximum and minimum stimulations are determined. Since, there are 16 Gabor filters, the index of each filter is an integer number between 0 and 15; therefore, a 4-bit binary code can be used to represent this number. Then, the binary indices of filters with the maximum and minimum stimulations are concatenated to produce an 8-bit binary code. In this way, each pixel of the image can be represented by an 8-bit code based on Gabor filters. As it is clear, the indices of two filters with maximum and minimum stimulations cannot be mathematically equal. Given this, corresponding 16 codes with equal indices for maximum and minimum stimulations never occurred. Accordingly, the number of possible patterns in this coding method is 2^8–16 = 240. For feature extraction, the input image is divided into several blocks; then, the histogram of these codes is calculated over each block using Eq. 2; the feature vector length for each block is 240. If we use 8 Gabor filters (instead of 16 filters), then each pixel can be represented by a 6-bit binary code. Consequently, feature vector length for each block of image is 2^6–2^3 = 56. An example of calculating this 6-bit binary code using 8 Gabor filters is shown in Fig. 5.

$$H_i = \sum_{x,y} I(P_I(x,y) = i) \;\; i \in \{\text{produced patterns}\} \tag{2}$$

Where P_I is the matrix of produced patterns using the proposed operator for input image, x and y are the coordinate of this matrix elements, and I function is defined as:

$$I(A) = \begin{cases} 1 \; A \; is \; true \\ 0 \; A \; is \; false \end{cases} \tag{3}$$

Feature vector for the proposed method is very shorter than concatenating all Gabor outputs in the feature vector. By concatenating all Gabor outputs, the feature vector length is equal to the product of the number of image pixels and the number of Gabor filters. For instance, for a

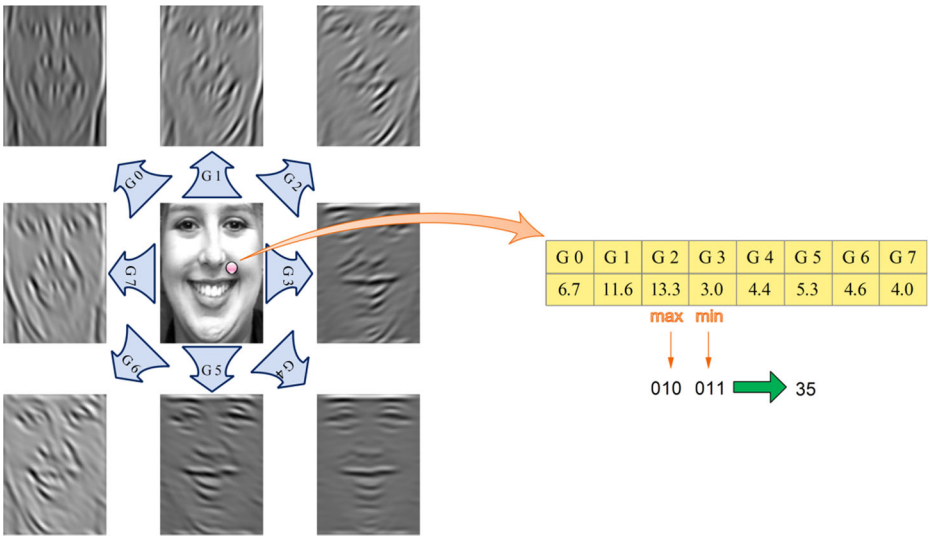

Fig. 5 An example of the proposed Gabor coding for a pixel of image

150 × 110 pixels image, the length of feature vector using 16 and 8 Gabor filters, are 264,000 (150 × 110 × 16) and 132,000 (150 × 110 × 8), respectively.

In computing the histogram of the produced patterns, each element of P_I can be considered with its Gabor responses in the histogram. In other words, each element is weighted using the minimum and maximum Gabor responses; then, the histogram of these patterns is computed over the weighted elements. In this weighing method, in addition to orientations of the selected filters, their response values containing important information of image edge intensities are included in the final histogram. For this purpose, Eq. 4 is utilized to compute the histogram of patterns instead of Eq. 2;

$$H_i = \sum_{x,y} M(x,y) \times I(P_I(x,y) = i) \quad i \in \{\text{produced patterns}\} \tag{4}$$

Where matrix M is calculated using the convolution results of Gabor filters and the input image via Eq. 5:

$$M(x,y) = \max(|\mathrm{G}_i * \mathrm{I}(\mathrm{x},\mathrm{y})|) - \min(|\mathrm{G}_i * \mathrm{I}(\mathrm{x},\mathrm{y})|) \quad i = 1, ..., n \tag{5}$$

where, G_i is the ith Gabor filter, I is input image, * denotes convolution operator, and n is the number of Gabor filters. Consequently, the difference of the maximum and the minimum responses is computed and included in the histogram. In this equation, the small values of $M(x, y)$ are due to small differences between the responses of Gabor filters. In other words, convolution of different Gabor filters and the input image at pixel (x, y) may not considerably differ from each other. Therefore, the image at pixel (x, y) is approximately a flat area and has not any important edge. Conversely, large value in matrix M indicates a strong edge with the same orientation as the filter with the maximum response. Hence, it can be concluded that the intensity of image edges are weighted in the computed histogram using Eq. 4.

To extract feature vector from the input image, facial region is cropped and normalized based on the preprocessing method introduced in Section 2.1. Then, the normalized image is divided into 7 × 6 blocks and the proposed feature is computed using 8 and 16 Gabor filters over these blocks. Finally, to create final feature vector, the histograms of these image blocks are concatenated into a single vector as shown in Fig. 6. Since preprocessing phase scales all facial images into a predefined size, a few fixed sizes Gabor filters can be employed for feature extraction.

2.3 Feature classification

To classify the extracted features based on the previous section, well-known Support Vector Machine (SVM) classifier is used. This classifier learns a hyperplane with maximum margin with all training data in different classes. The hyperplane is obtained by optimization on training data. In test phase, test sample x is classified (Eq. 6) based on the achieved support vectors in training phase as:

$$f(x) = \sum_{i=1}^{L} \alpha_i y_i k(x_i - x) + b \tag{6}$$

where, x_i and y_i are support vectors and their labels, respectively; k is kernel function of SVM classifier; parameters α and β are obtained in the training phase. In this paper, linear and polynomial (with degree of 3) functions are used as SVM kernel functions. Originally, SVM is

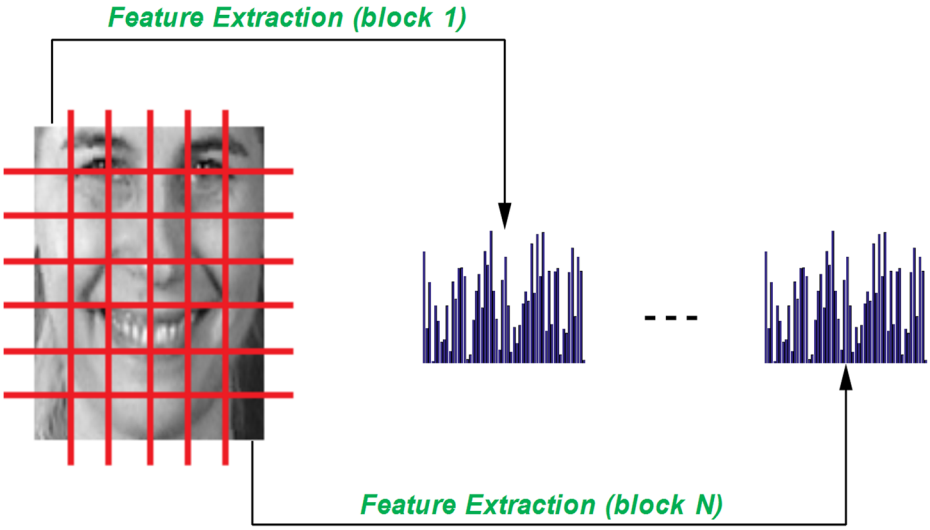

Fig. 6 Blocking the normalized facial image and feature extraction from these blocks. Final feature vector is achieved by concatenating these block features

a binary (two-class) classifier; here, one-versus-all method is utilized to extend original SVM into a multi-class classifier for facial expression recognition. To implement SVM classifier, LIBSVM toolbox [4] is employed.

3 Evaluation of the proposed method

In this section, facial expression datasets which are used in the experiments are first introduced. Then, the evaluation method and the obtained results of the proposed algorithm are compared with some other methods.

3.1 Datasets

To evaluate the proposed method, CK+, SFEW, and MMI facial expression recognition datasets are used; these datasets are briefly described as bellow:

Extended Cohn-Kanade dataset (CK+): CK+ is one of the comprehensive and well-known datasets in facial expression and action unit analysis [39]. This dataset contains 593 image sequences (with different numbers of images) taken from 123 subjects with 327 sequences labeled as a facial expression (including six basic emotions and contempt). In CK+, each image sequence starts from neutral emotion and reaches to its peak one in the last frame. In this paper, the last frame of these 327 image sequences are chosen for experiments; some samples of these frames are selected and shown in Fig. 7. The number of image sequences for anger, contempt, disgust, fear, happiness, sadness, and surprise expressions in CK+ dataset are 45, 18, 59, 25, 69, 28, and 83, respectively.

Static Facial Expression in the Wild (SFEW): Uncontrolled SFEW 2.0 dataset, which is very challenging due to its large variations in illumination, head pose, etc., is introduced

in EmotiW 2015 challenge [7]. SFEW 2.0 contains train, validation, and test image sets with 958, 436, and 372 images, respectively. In this dataset, the expression labels of train and validation sets are publicly available; however, for challenging purposes, test set labels are not published. Because of this, in this study, the training and validation sets are utilized for training and test phases, respectively. Some example images of uncontrolled SFEW dataset are shown in Fig. 8.

MMI: The extended version of original MMI dataset [48] contains 213 video clips from 20 European, Asian, and South American subjects [57]. In this dataset, there are spontaneous facial expressions; moreover, a number of subjects have glasses making it more challenging to recognize facial expressions. In MMI dataset, each video with the resolution of 720 × 576 pixel starts from neutral image and reaches to the peak of emotion and returns to the neutral one. In this paper, two frames (including the peak and the neutral ones) of each video labeled as one of the six basic emotions are chosen and used in experiments. MMI dataset has only 6 basic-emotions, and does not provide separate data for the neutral. In this paper, to increase the number of classes/facial expressions, and to make the experiments more challenging, the neutral frame of each video in MMI dataset is selected as a sample of neutral class. The experiments are performed on the neutral and peak frames independently as different classes; in other words, in the proposed algorithm, only one image is needed to recognize the facial expression, and there is no need for two frames. Some sample frames from MMI dataset are shown in Fig. 9.

RAF-DB: Real-world Affective Faces (RAF) database [57] includes 29,672 uncontrolled images downloaded from Internet websites. The facial images in RAF-DB are labeled 40 times by 315 learned annotators and the noisy labels are removed via Expectation Maximization. The uncontrolled images in RAF-DB are provided in two sets including the basic emotions and compound ones. In addition to the original images, face bounding boxes, 5 manually annotated facial landmarks, 37 automatic landmarks, and aligned facial images are provided in this dataset. The set of basic emotions consists of 15,339 facial images (labeled as one of six basic emotions plus neutral one) in training and test sub-sets with 12,271 and 3068 images, respectively. Some aligned facial images from RAF-DB are shown in Fig. 10. In this paper, the basic emotions-set of RAF-DB is used for the experiments.

3.2 Experiments

This section describes the experiments of the proposed facial expression recognition system on the aforementioned datasets. The proposed method using 16 Gabor filters ($v = 1$ and 2), 7 × 6 image blocks, and SVM classifier (with linear kernel) is applied on CK+ and MMI datasets. Confusion matrices of facial expression recognition on CK+ and MMI datasets are shown in Tables 1 and 2, respectively. In these experiments, a person-independent (PI) 10-fold cross validation train-test scheme is employed; in this evaluation, data are randomly divided into 10 almost equal subsets. One subset is used as test data and the remaining 9 subsets are considered as training data. In this method which is more challenging than person-dependent (PD) experiment, there is not any image from training persons in the test data. This process is repeated 10 times, so that each of these 10 subsets is considered as test data. Finally, recognition rate is reported based on the test results of all images. According to Table 1,

Fig. 7 Some sample images of CK+ dataset [39]

recognition rates for contempt, fear, and sadness expressions are less than 90%; however, it is more than 90% for the other ones. With regard to CK+ dataset, it can be seen that the number of images for contempt, fear, and sadness expressions are considerably less than the others. Inadequate data in these expressions affect the training phase and reduce the accuracy of classification. On the other hand, as can be seen from Table 2, recognition rates for MMI dataset are less than CK+ due to its spontaneous expressions. The achieved average recognition rates for CK+ and MMI datasets are 95.11 and 71.15%, respectively.

The parameters of Gabor filters can be tuned via sampling and search strategy based on Information Diagram [42]. However, this study does not focus on Gabor parameters tuning. In order to explore the effect of Gabor filter size v (frequency) on recognition rate, different sizes are used for feature extraction via 8 filters scenario. The extracted features are classified via linear kernel SVM classifier. The recognition rates with $v = 1$ to 5 for CK+ dataset are shown in Fig. 11. As can be seen from this figure, increasing the size of Gabor filters up to $v = 3$ improves the recognition rate. However, more increasing the size of Gabor filters ($v = 4$ and 5) reduces the recognition rate. Proper frequency of Gabor filters is depend on the scale of facial images; here, the facial images are re-scaled into 150×110 and $v = 3$ yields the highest recognition rate. For the larger (or smaller) image scales, Gabor filters with larger (or smaller) v can be more appropriate to accurately represent expression-related texture information of facial components.

Fig. 8 Some sample images of uncontrolled SFEW dataset [7]

Fig. 9 Some sample images of MMI dataset

To compare the proposed method with other image texture description method, some of the most important feature extraction methods including LBP, HOG, POEM, LDN, DCP, HWP, CBFD, and CA-LBFL along with the proposed one (using 8 and 16 Gabor filters) are evaluated on CK+, MMI, and SFEW datasets. The achieved results of the proposed method are compared with the other texture descriptors in Table 3. The parameters of the proposed method are the number of image blocks, the frequency and the number of Gabor filters, and classifier parameters. With regard to the number of image blocks, facial image (without any parameter-tuning) is divided into 7×6 blocks according to the proposed method in [54]. The numbers of Gabor filters are 8 and 16 and their frequencies are selected as $v = 2$ for 8 Gabor filters and $v = 2$ and 3 for 16 Gabor filters based on the previous experiments shown in Fig. 11. The classifier used in these experiments is well-known SVM classifier (with linear kernel without any parameter). Increasing (or decreasing) the number of image blocks increases (or decreases) the accuracy of all feature descriptor methods; hence, for a fair comparison, (according to the proposed method in [54]) facial image is divided into 7×6 blocks for all features except HOG which is not a large-block-based descriptor. Since the performances of the compared descriptors were reported in different applications via various classifiers, here, the same as the proposed method, they are experimented using SVM classifier (with linear kernel) on the facial expression recognition datasets. In these experiments, the parameters of the compared descriptors are set as proposed values by their publishers. Specially, to extract LBP features, the histograms of uniform LBPs (with 8 neighborhoods and radius 1) of these blocks are concatenated into a single feature vector. The length of uniform LBP histogram for each block of image is 59; these parameters are selected similar to [54]. In LDN descriptor, the histogram length for each block is 56 as [49]. For POEM, uniform LBPs with 8 neighborhoods and radius 1, and 4-bin histogram of gradients over 0 to 180 degree are used. In HOG feature extraction, 10×10 (pixels) cell size, 16-bin histogram of gradients over 0 to 180 degree are utilized. In DCP and HWT, R1 and R2 are set to 4 and 6 as [3], respectively. The parameters of CA-LBFL and CBFD are set as described in [9, 38], respectively. The process of these experiments for CK+ and MMI datasets are similar to the performed ones in Tables 1 and 2, respectively. In uncontrolled SFEW dataset, training and validation sets are provided separately; in our experiments, these sets are used as training and testing data, respectively. Similarly, uncontrolled RAF-DB provided separate training and test sets which are used with the same purposes in the experiments. For CK+ and MMI datasets, the cross-validation testing scheme is repeated ten times and average recognition rates along with 95% confidence

Fig. 10 Some aligned facial images of RAF database

Table 1 Confusion matrix of facial expression recognition for CK+ dataset

	Angry	Contempt	Disgust	Fear	Happiness	Sadness	Surprise
Angry	**91.11**	0	4.44	0	0	4.44	0
Contempt	5.55	**88.88**	0	0	0	5.55	0
Disgust	1.69	0	**94.91**	0	0	1.69	1.69
Fear	0	0	0	**88**	8	4	0
Happiness	0	0	0	0	**100**	0	0
Sadness	3.57	3.57	0	0	0	**89.28**	3.57
Surprise	0	1.2	0	0	0	0	**98.79**
						Average:	**95.11**

The bold entries show the highest values in each row

intervals [35] are reported in Table 3. In SFEW and RAF-DB, since the test sets are fixed and provided by dataset publishers, their test sets are split into 10 sub-sets in order to calculate 95% confidence interval of recognition rates. As can be seen from this table, the recognition rates of the proposed method are higher than the other ones.

For feature extraction from a 150 × 110 pixels image via 8 Gabor filters, first, the filters are convolved with the input image. The output size of this step is 132,000 (150 × 110 × 8). In other words, there are 8 different values for each pixel (or 8 matrices of 150 × 110 for the input image). In each pixel, 8 complex values are then converted to a decimal number as described in Section 2.2.2 (Fig. 5). The output size of this step is 16,500 (150 × 110) which is 8 times smaller than the previous step. To further reduce the dimension of features, weighted histograms of these decimal numbers are calculated over different parts of image as facial feature vector. The length of these histograms using 8 Gabor filters are 56 (calculation of the histogram is described in Section 2.2.2 and Eq. 4). If the image is divided into 42 sub-regions, the length of final feature vector, which is obtained by concatenating the histograms of sub-regions, is 2352 (42 × 56).

In Table 4, recognition rates of the proposed feature extraction method using 16 Gabor filters (with $v = 2$ and 3), 7 × 6 image blocks, and SVM classifier (polynomial kernel with degree 3) are compared with the reported recognition rates of some recent facial expression recognition methods. In these experiments, a 10-fold cross validation testing scheme is employed for CK+ and MMI datasets; for SFEW dataset, train and test sets are chosen according to the reported (train and validation) ones in the dataset. The number of facial expressions (classes) and experimental person dependencies (person independent (PI)/person dependent (PD)) of these methods are shown in the table. In Table 4, to avoid any bias in execution and inappropriate parameter settings of other methods, the published reported values

Table 2 Confusion matrix of facial expression recognition for MMI dataset

	Angry	Disgust	Fear	Happiness	Sadness	Surprise	Neutral
Angry	**63.63**	12.12	3.03	6.06	6.06	0	9.09
Disgust	6.25	**65.62**	3.12	9.37	6.25	3.12	6.25
Fear	7.14	3.57	**60.71**	3.57	0	10.71	14.28
Happiness	0	2.38	0	**78.57**	0	2.38	16.66
Sadness	3.12	0	6.25	3.12	**59.37**	3.12	25
Surprise	0	0	7.31	0	4.87	**70.73**	17.07
Neutral	3.36	2.40	6.73	3.84	4.32	4.32	**75**
						Average:	**71.15**

The bold entries show the highest values in each row

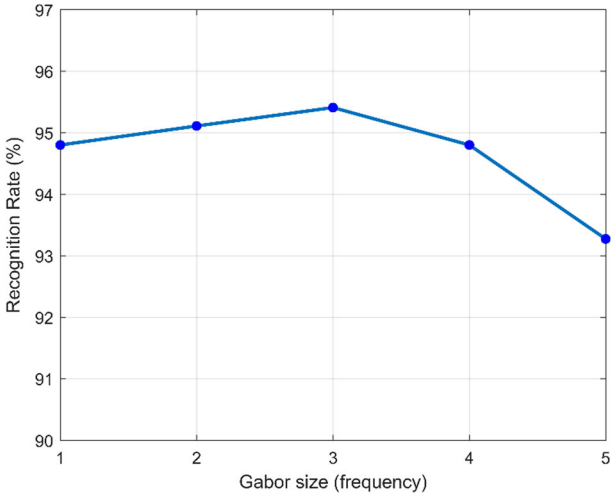

Fig. 11 Recognition rate of the proposed feature extraction method using linear kernel SVM classifier and 8 Gabor filters with different frequencies ($v = 1$–5)

in the compared methods are utilized. Therefore, for each work, only the datasets tested and reported in the respective publications were compared and shown in this table. According to this table, the proposed method has acceptable facial expression recognition rates on both controlled and uncontrolled datasets compared to the other method.

4 Conclusion

In this study, inspired by human vision system, a new image feature extraction method is introduced based on Gabor filters. In this method, each Gabor filter is considered as a vision cell and is convolved with input image. In each pixel of the image, Gabor filters with the maximum and minimum stimulations are selected and coded via a novel coding method. Then, this code is weighted using the responses of selected two filters. Finally, histogram of the

Table 3 Facial expression recognition using the proposed method on CK+, SFEW, and MMI datasets compared to the other feature extraction methods

Feature	Feature length	Dataset			
		CK+	SFEW	MMI	RAF-DB
LBP [54]	2478	92.66 ± 0.75	27.98 ± 2.18	58.06 ± 1.14	57.24 ± 0.92
HOG [55]	2640	93.88 ± 0.65	25.69 ± 2.67	69.19 ± 1.35	66.95 ± 1
POEM [58]	9912	94.19 ± 0.88	30.05 ± 2.05	68.26 ± 0.8	69.01 ± 0.89
LDN [49]	2352	91.13 ± 0.91	18.35 ± 2.67	58.29 ± 0.93	57.33 ± 1.48
DCP [8]	21,504	90.21 ± 0.97	28.67 ± 2.44	64.45 ± 1.19	63.46 ± 0.71
HWP [3]	21,504	91.44 ± 1.43	29.36 ± 2.16	64.93 ± 0.7	65.74 ± 0.79
CBFD [38]	21,000	93.27 ± 0.64	34.63 ± 1.27	68.96 ± 0.94	72.61 ± 0.98
CA-LBFL [9]	21,000	94.19 ± 0.72	34.86 ± 1.73	68.72 ± 0.93	74.22 ± 0.59
Proposed method (8 Gabor filters)	2352	95.11 ± 0.39	36.01 ± 1.25	69.95 ± 0.55	75.10 ± 0.59
Proposed method (16 Gabor filters)	10,080	**95.72 ± 0.21**	**36.92 ± 0.74**	**71.88 ± 0.56**	**76.23 ± 0.49**

The bold entries show the highest recognition rates for each dataset

Table 4 Comparison of the proposed method with the other facial expression recognition methods

Feature	Experiment		Dataset			
	Classes	PD/PI	CK+	SFEW	MMI	RAF-DB
TPOEM [5]	7	PI	92.91	–	–	–
[19]	6	PD	94.09	–	–	–
PHOG + LPQ [7]	7	PI	–	35.93	–	–
RVLBP + SVM [64]	7	PI	–	38.07	–	–
DS-GPLVM [11]	7	PI	–	18.81	–	–
STM-ExpLet [36]	6	PI	94.19	30.03	75.12	–
[44]	6	PI	90	–	–	–
[34]	7	PD	86.58	–	–	–
LBP + NCM [15]	7	PD	91.95	–	–	–
DLP-CNN + SVM [33]	7	PI	–	–	–	74.2
Boosting-POOF [37]	7	PI	–	–	–	73.19
Proposed method (16 Gabor filters)	7	PI	**96.33 ± 0.19**	**39.68 ± 0.75**	**77.4 ± 0.6**	**76.23 ± 0.42**

The bold entries show the highest recognition rates for each dataset

weighed codes is computed over the whole normalized facial image. Thus, the proposed method utilizes both Gabor convolution responses and filter orientations in a histogram-based feature vector. Experimental results on three different datasets (including CK+, MMI, and SFEW) in both controlled and uncontrolled conditions show the effectiveness of the proposed method compared to the existing appearance feature-based facial expression recognition methods. Moreover, the proposed feature extraction method outperforms the state-of-the-art texture descriptors on these datasets.

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